

Scientific Machine Learning for Basic Sciences

(BSc Physics | BSc Maths) — A 6-Month Research & Industry-Ready Certificate Program

Duration: 6 Months | Total: 180+ Hours | Format: Theory + Computational Labs + Capstone Project
Target Audience: B.Sc / M.Sc Students in Physics, Mathematics, Statistics, Chemistry & Computational Engineering
Core Stack: Python, NumPy, SciPy, PyTorch, PINNs, Autograd, ODEs/PDEs, DeepONet, FNO, Google Colab

Program Overview

This program is specifically designed for students in basic sciences—particularly **Physics and Mathematics**—to bridge the gap between fundamental physical principles and modern artificial intelligence. Rather than treating AI as a black box, students learn how to integrate scientific physical laws, differential equations, and conservation principles directly into neural network architectures using **Physics-Informed Neural Networks (PINNs)** and **Neural Operators**.

Students progress methodically from mathematical foundations and scientific computing in Python, through classical numerical methods and machine learning, into differentiable computing with PyTorch autograd, forward/inverse PINNs, and advanced research capstones. Every topic follows the rigorous cycle: *Physical Law* → *Mathematical Formulation* → *Numerical Solver* → *Differentiable AI* → *Empirical Validation*.

Curriculum Breakdown (6 Comprehensive Modules)

Module	Core Focus & Competencies	Key Topics & Theory	Hands-on Laboratory Deliverable
Module 01 Foundations	Mathematical & Computational Foundations for AI	- Python fundamentals, data structures & scientific Git workflows - Linear algebra: vectors, matrices, eigenvalues & eigenvectors - Multivariable calculus: partial derivatives, gradients, Jacobian & Hessian - Probability, uncertainty quantification & numerical error basics	Simulate a noisy physical dataset, implement linear regression from scratch using gradient descent, and publish a reproducible Python notebook.
Module 02 Numerical Computing	Scientific Computing, Numerical Methods & Differential Equations	- NumPy vectorization, SciPy routines & data visualization - Numerical errors, stability, convergence & interpolation - Numerical ODE solvers: Euler, Runge–Kutta (RK45) - PDE fundamentals: 1D heat equation, wave equation & finite-difference schemes	Implement a 1D finite-difference heat-equation solver, generate reference simulation datasets, and benchmark numerical convergence across grid resolutions.
Module 03 Scientific ML	Machine Learning for Scientific & Engineering Applications	- Regression models: regularized linear regression (Lasso/Ridge), polynomial regression - Classification: logistic regression, decision trees, Random Forest - Model evaluation: bias-variance trade-off, cross-validation & hyperparameter tuning - Dimensionality reduction (PCA), feature scaling & scientific baselines	Build an end-to-end scientific ML pipeline predicting material properties from experimental measurements; deliver an error analysis report.
Module 04 Deep Learning	Deep Learning, PyTorch & Differentiable Computing	- Neural network architectures: perceptrons, activation functions & backpropagation - Automatic differentiation: PyTorch autograd engine & gradient graphs - CNNs for spatial physical fields and scientific image data - Optimization: SGD, Adam, learning rate schedules, and GPU acceleration	Train a deep neural network to approximate complex non-linear functions; compute high-order analytical derivatives using PyTorch autograd.

Module 05 SciML & PINNs	Scientific Machine Learning & Physics-Informed Neural Networks (PINNs)	• The SciML paradigm: data-driven vs physics-constrained models • PINN formulation: PDE loss residuals, boundary/initial conditions & collocation points • Solving forward physical problems: harmonic oscillators and 1D Burgers' equation • Inverse PINNs: parameter discovery (recovering unknown viscosity/damping from noisy data)	Formulate and train a PINN for a non-linear differential equation; compare solution accuracy and speed against classical numerical solvers (SciPy/RK45).
Module 06 Research Capstone	Advanced Scientific AI, Research Engineering & Industry Capstone	• Advanced PINN training: adaptive collocation sampling, loss balancing (gradNorm) • Neural operators: DeepONet and Fourier Neural Operators (FNO) concepts • Research engineering: experiment tracking, configuration management & modular code • Scientific technical writing, reproducibility, publication-ready plots & oral viva defense	Final Capstone Project: End-to-end scientific AI research project with a GitHub repository, benchmark report, poster presentation, and formal project defense.

Capstone Defense & Deliverables

Every student must complete and defend an independent or paired capstone research project before a review panel. Example capstone themes include:

- **Physics-Informed Fluid Dynamics:** Solving 1D Burgers' and Navier-Stokes approximations with PINNs.
- **Inverse Parameter Identification:** Estimating unknown physical constants from sparse sensor observations.
- **Neural Operator Benchmarking:** Training Fourier Neural Operators (FNO) to solve parametric partial differential equations.
- **Quantum & Chemical Kinetics:** Simulating reaction rates and molecular energy surfaces using physics-guided ML.

Career Pathways & Dual Certification

Upon successful completion and defense of the capstone project, students receive an **Industry & Research Verified Certificate** with unique QR verification. Graduates are equipped for roles including **Scientific AI Researcher**, **Computational Data Scientist**, **SciML / PINNs Engineer**, **Simulation Specialist**, and **Quantitative Analyst** in premier research institutions, aerospace, energy, semiconductor, and quantitative finance sectors.